

AI3602 Data Mining Homework 5

Kailing Wang 521030910356

June 1.2024

Question 1

(1) To achieve ϵ -differential privacy, we choose the scale parameter b as

$$b = \frac{\Delta f}{\epsilon}, \quad (1)$$

where

$$\Delta f = \max_{|D_1 \Delta D_2|=1} \|f(D_1) - f(D_2)\|_1. \quad (2)$$

Oh, here given L1 sensitivity, is 1 so $b = \frac{1}{\epsilon}$.

(2)

$$\sigma = \frac{\Delta f \sqrt{2 \ln(1.25/\delta)}}{\epsilon} \quad (3)$$

where

$$\Delta f = \max_{|D_1 \Delta D_2|=1} \|f(D_1) - f(D_2)\|_2. \quad (4)$$

So $\sigma = \frac{\sqrt{2 \ln(1.25/\delta)}}{\epsilon}$. Note that I assume the σ in the question should be σ^2 as normal representation.

Question 2

To satisfy ϵ -differential privacy, the following condition must hold:

$$e^\epsilon \geq \frac{p}{\frac{1-p}{m-1}} \quad (5)$$

$$e^\epsilon(1-p) \geq p(m-1) \quad (6)$$

$$e^\epsilon - e^\epsilon p \geq p(m-1) \quad (7)$$

$$e^\epsilon \geq p(m-1 + e^\epsilon) \quad (8)$$

Thus, the probability p that satisfies ϵ -differential privacy is:

$$p \leq \frac{e^\epsilon}{(m-1) + e^\epsilon} \quad (9)$$

To satisfy ϵ -differential privacy when y is not the correct answer, the following condition must hold:

$$e^\epsilon \geq \frac{\frac{1-p}{m-1}}{p} \quad (10)$$

$$e^\epsilon p \geq \frac{1-p}{m-1} \quad (11)$$

$$e^\epsilon p(m-1) \geq 1-p \quad (12)$$

$$e^\epsilon p(m-1) + p \geq 1 \quad (13)$$

$$p(e^\epsilon(m-1) + 1) \geq 1 \quad (14)$$

Thus, the probability p that satisfies ϵ -differential privacy is:

$$p \geq \frac{1}{(m-1)e^\epsilon + 1} \quad (15)$$

Combining both conditions, the probability p must satisfy:

$$\frac{1}{(m-1)e^\epsilon + 1} \leq p \leq \frac{e^\epsilon}{(m-1) + e^\epsilon} \quad (16)$$

Original Questions

Reference

lecture notes

1.
 - (1). Calculate b from the Laplace distribution $Lap(x|b)$ that satisfies ϵ -differential privacy with an L1-sensitivity of 1.
 - (2). Calculate σ from the Gaussian distribution $\mathcal{N}(0,\sigma)$ that satisfying (ϵ,δ) -differential privacy with an L2-sensitivity of 1. (No need to prove)
2. (Generalized Random Response). Consider a survey consisting of m potential candidate answers ($m \geq 2$). An individual participant sends the genuine answer with probability p or gives an arbitrarily other answer with probability $(1-p)/(m-1)$. Determine the probability p in order to satisfy ϵ -differential privacy.

Figure 1: Question 1